

(58) **Field of Classification Search**

CPC H02M 3/33569; H02M 3/33571; H02M
1/0003; H02M 1/0012; H02M 1/0016;
H02M 1/0025; H02M 3/07; H02M 3/075
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,782,027 B2 8/2010 Williams
7,786,712 B2 8/2010 Williams
7,812,579 B2 10/2010 Williams
7,977,927 B2 7/2011 Williams
8,212,541 B2 7/2012 Perreault et al.
8,824,978 B2 9/2014 Briffa et al.
8,829,993 B2 9/2014 Briffa et al.
8,830,709 B2 9/2014 Perreault et al.
8,830,710 B2 9/2014 Perreault et al.
8,860,396 B2 10/2014 Giuliano
8,957,727 B2 2/2015 Dawson et al.
9,020,453 B2 4/2015 Briffa et al.
9,160,287 B2 10/2015 Briffa et al.
9,166,536 B2 10/2015 Briffa et al.
9,172,336 B2 10/2015 Briffa et al.
9,209,758 B2 12/2015 Briffa et al.
9,490,752 B2 11/2016 Briffa et al.
9,531,291 B2 12/2016 Perreault
9,537,456 B2 1/2017 Briffa et al.
9,634,577 B2 4/2017 Perreault
9,755,672 B2 9/2017 Perreault et al.
9,768,731 B2 9/2017 Perreault et al.
9,768,732 B2 9/2017 Briffa et al.
9,917,517 B1 3/2018 Jiang et al.
9,979,421 B2 5/2018 Astrom et al.
9,991,913 B1 6/2018 Dinur et al.
10,164,577 B2 12/2018 Briffa et al.
10,193,441 B2 1/2019 Giuliano
10,218,268 B1* 2/2019 Kamble H02M 3/073
10,291,184 B1 5/2019 Sutskover et al.
10,389,235 B2 8/2019 Giuliano
10,658,981 B2 5/2020 Briffa et al.
10,749,476 B2 8/2020 Sutskover et al.
10,862,428 B2 12/2020 Henzler et al.
10,992,265 B2 4/2021 Hoversten et al.
11,031,864 B2 6/2021 Low et al.
11,043,918 B2 6/2021 Tanaka et al.
11,191,028 B2 11/2021 Hoversten et al.
11,245,367 B2 2/2022 Garrett et al.
11,558,014 B2 1/2023 Tanaka et al.
11,637,531 B1 4/2023 Perreault et al.

11,664,727 B2 5/2023 Giuliano et al.
11,736,010 B2 8/2023 Giuliano et al.
11,757,410 B2 9/2023 Hoversten et al.
11,757,411 B2 9/2023 Tanaka et al.
11,764,669 B2 9/2023 Chen et al.
2006/0176098 A1* 8/2006 Chen H02M 3/156
327/341
2009/0212830 A1* 8/2009 Zamprogno H03F 3/45977
327/108
2011/0012765 A1* 1/2011 Hsieh H03K 5/249
341/143
2013/0113550 A1* 5/2013 Bawa H03H 19/004
327/554
2018/0375434 A1 12/2018 Biziitu et al.
2020/0099290 A1* 3/2020 Szczeszynski H02M 3/07
2020/0228016 A1 7/2020 Wu et al.
2021/0288614 A1 9/2021 Hoversten et al.
2021/0385752 A1 12/2021 Hoversten et al.
2022/0103066 A1 3/2022 Chen et al.
2022/0131463 A1 4/2022 Giuliano et al.
2022/0149725 A1 5/2022 Garrett et al.
2023/0054485 A1 2/2023 Hoversten et al.
2023/0057037 A1 2/2023 Hoversten et al.

OTHER PUBLICATIONS

Cao, et al.; "Multiphase Multilevel Modulator DC-DC Converter for High-Current High-Gain TEG Application"; IEEE Transactions on Industry Application; vol. 47; No. 3; May/Jun. 2011; 9 Pages.
Chen, et al.; "A High Efficiency High Power Step-Up Resonant Switched-Capacitor"; 2012 IEEE Energy Conversion Congress and Exposition (ECCE); Sep. 15-20, 2012; 5 Pages.
Shoyama, et al.; "Resonant Switched Capacitor Converter with High Efficiency"; 2004 35th Annual IEEE Power Electronics Specialists Conference; 2004; 7 Pages.
Sun, et al.; "High Power Density, High Efficiency System Two-Stage Power Architecture for Laptop Computers"; 2006 37th IEEE Power Electronics Specialists Conference; Jun. 18-22, 2006; 7 Pages.
Unknown; "Fundamentals of Power Electronics"; Chapter 19: Resonant Conversion; Sep. 19, 2019; 87 Pages.
Yeung, et al.; "Multiple and Fractional Voltage Conversion Ratios for Switched-capacitor Resonant Converters"; 2001 IEEE 32nd Annual Power Electronics Specialists Conference (IEEE Cat. No. 01CH37230); Jun. 17-21, 2001; 6 Pages.
PCT International Search Report and Written Opinion dated Jan. 10, 2023 for International Application No. PCT/US2022/041011; 12 pages.

* cited by examiner

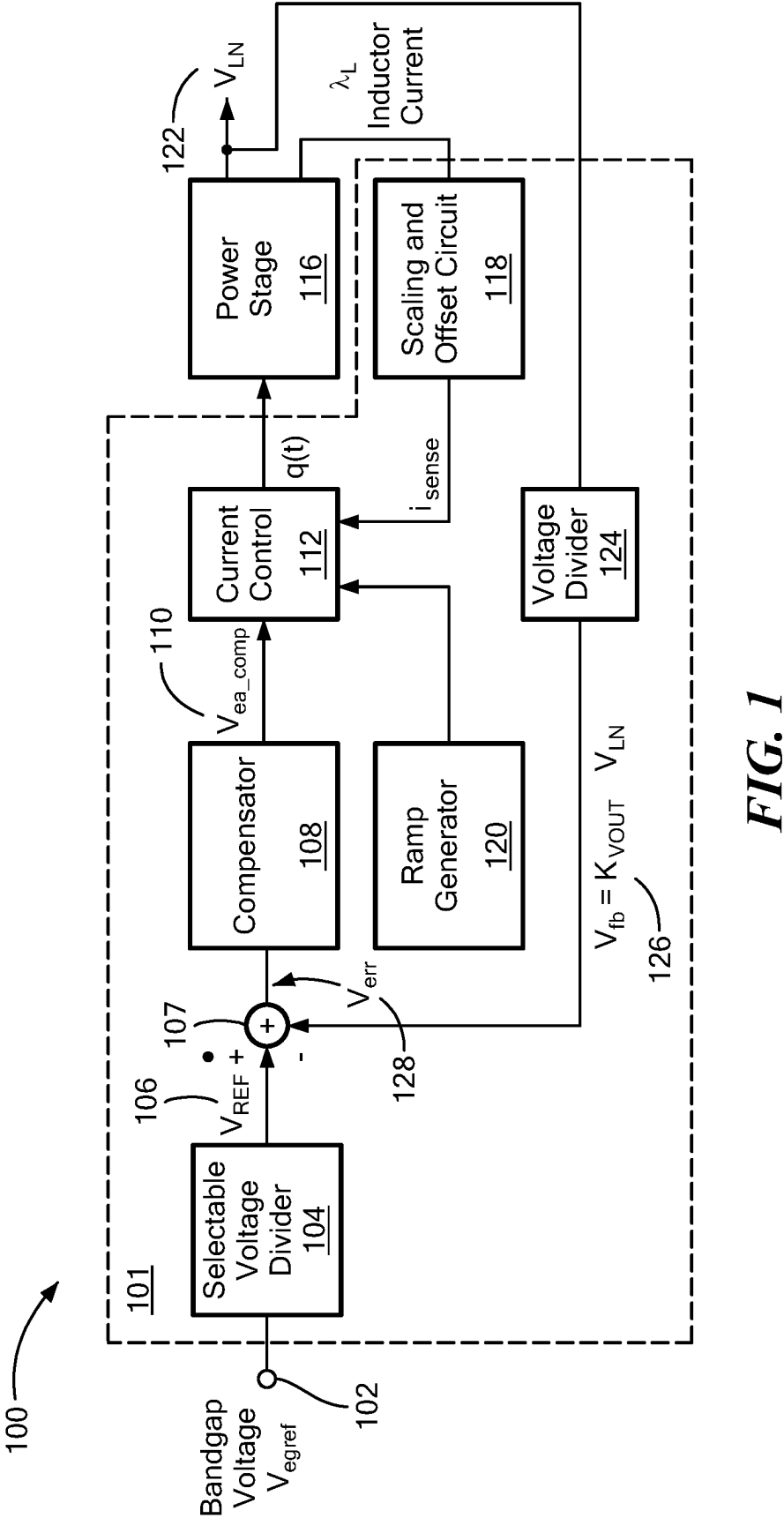


FIG. 1

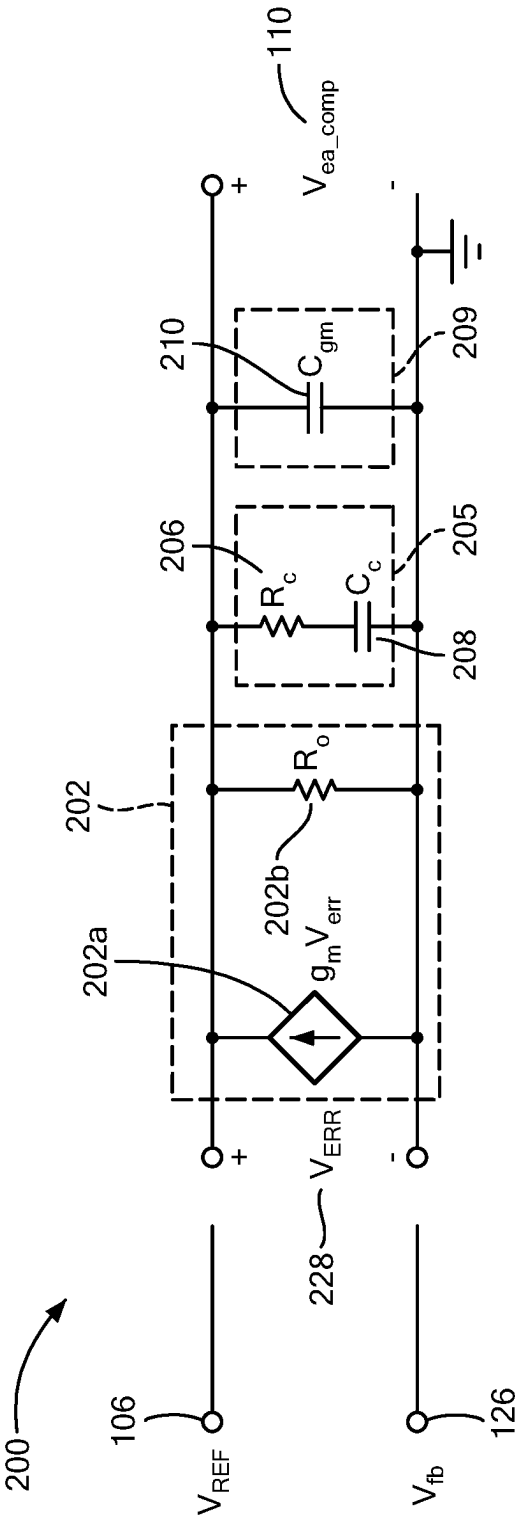


FIG. 2
Prior Art

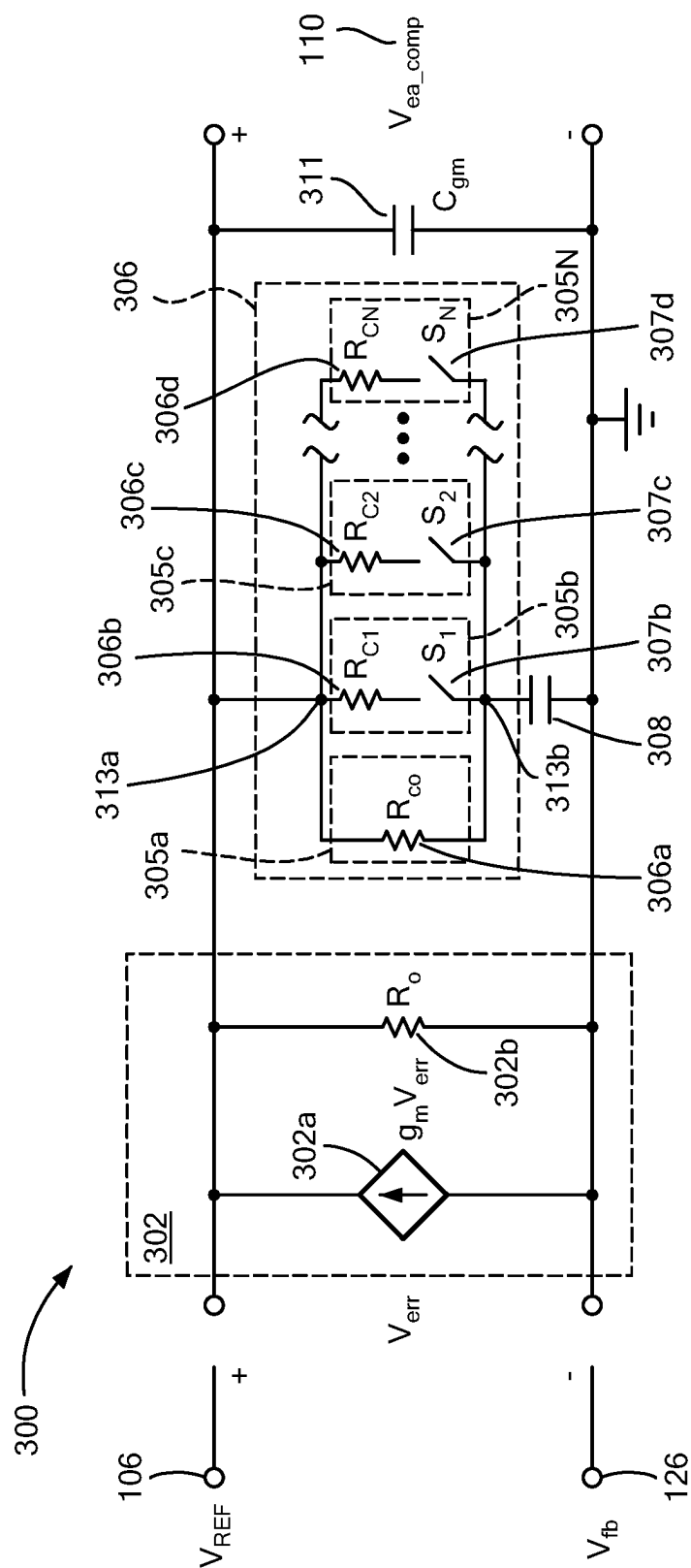


FIG. 3

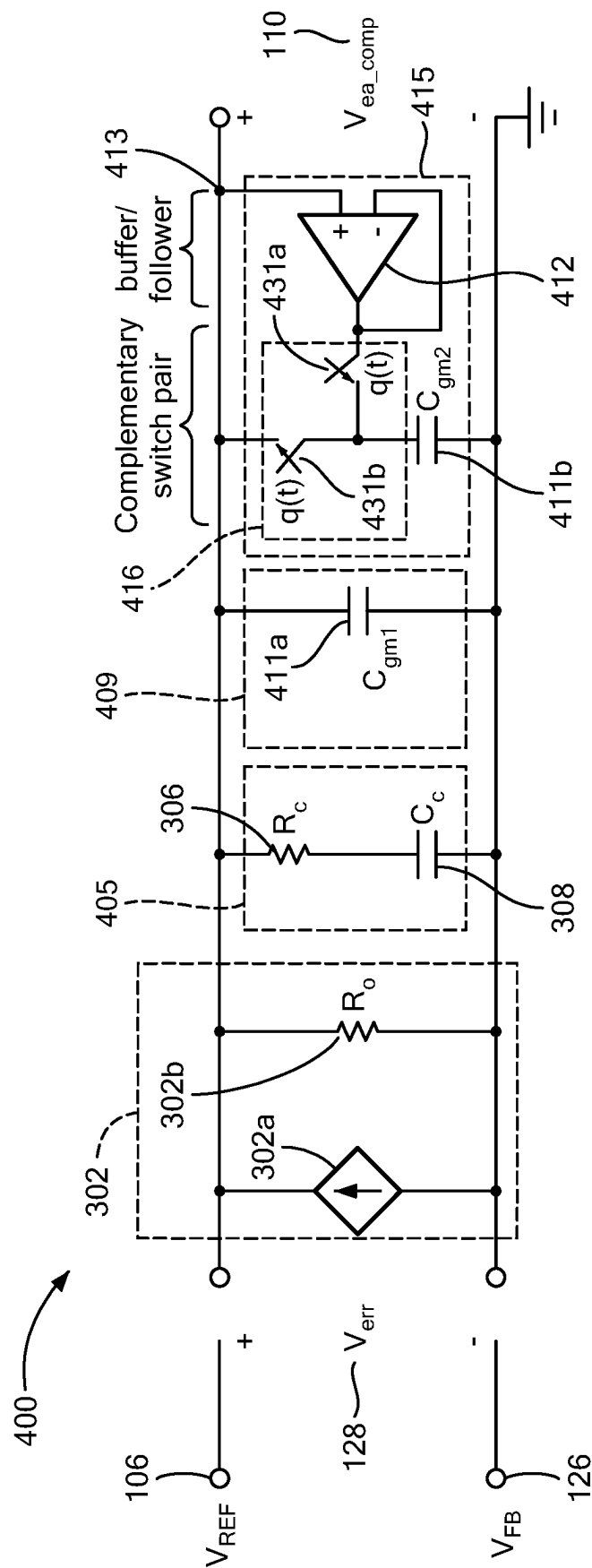


FIG. 4

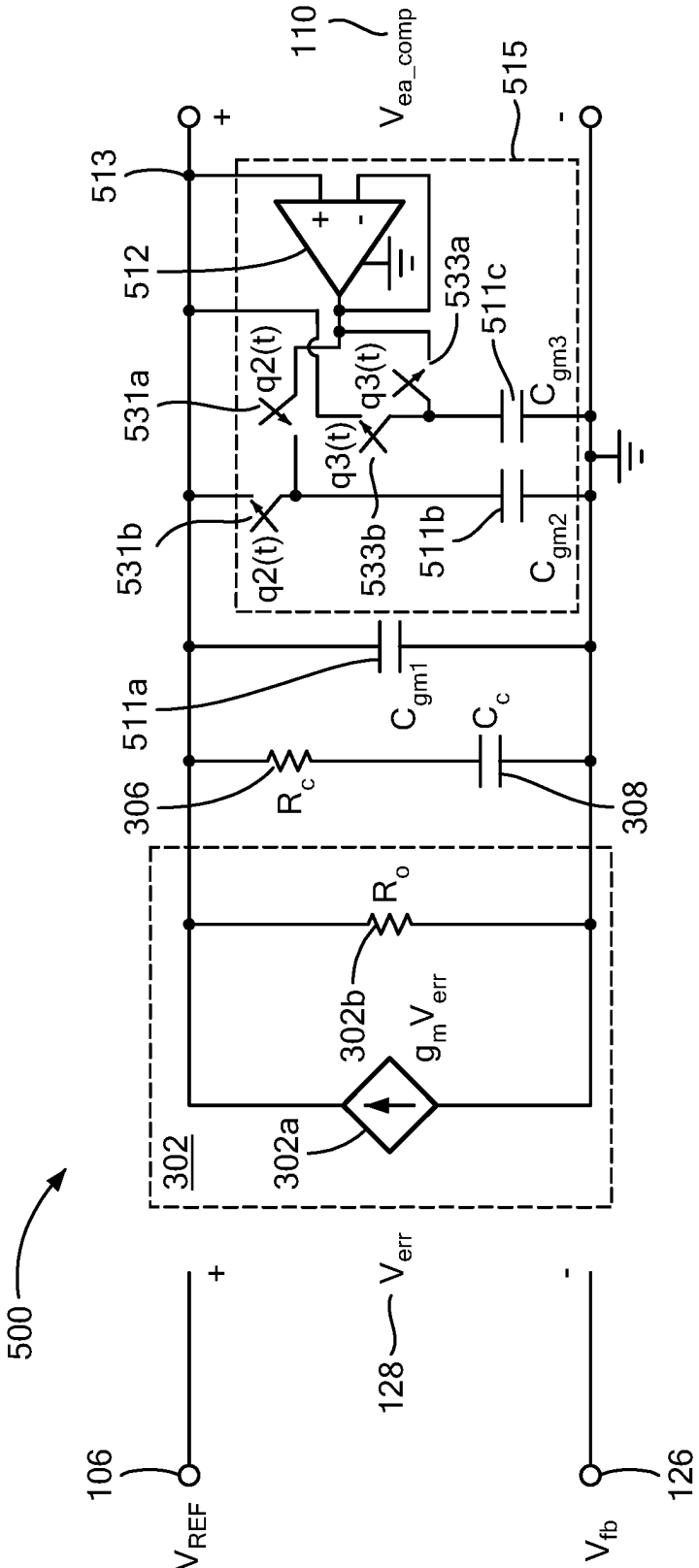


FIG. 5

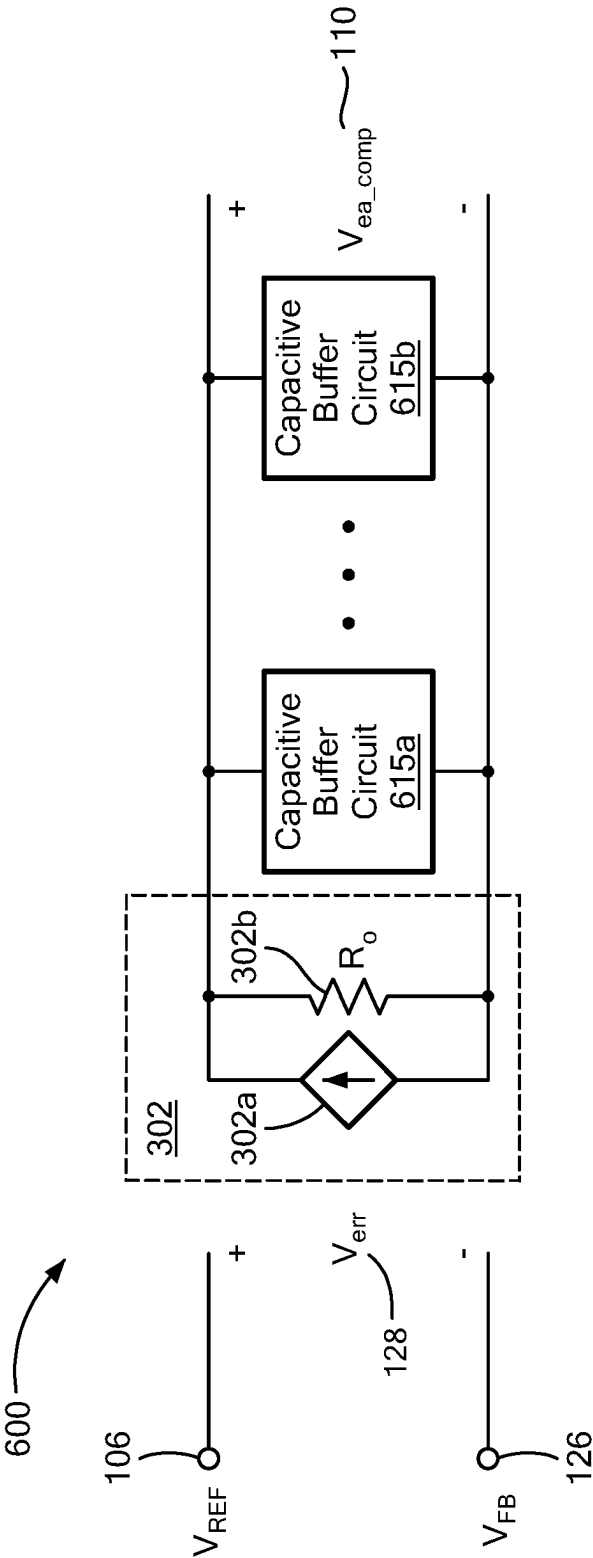
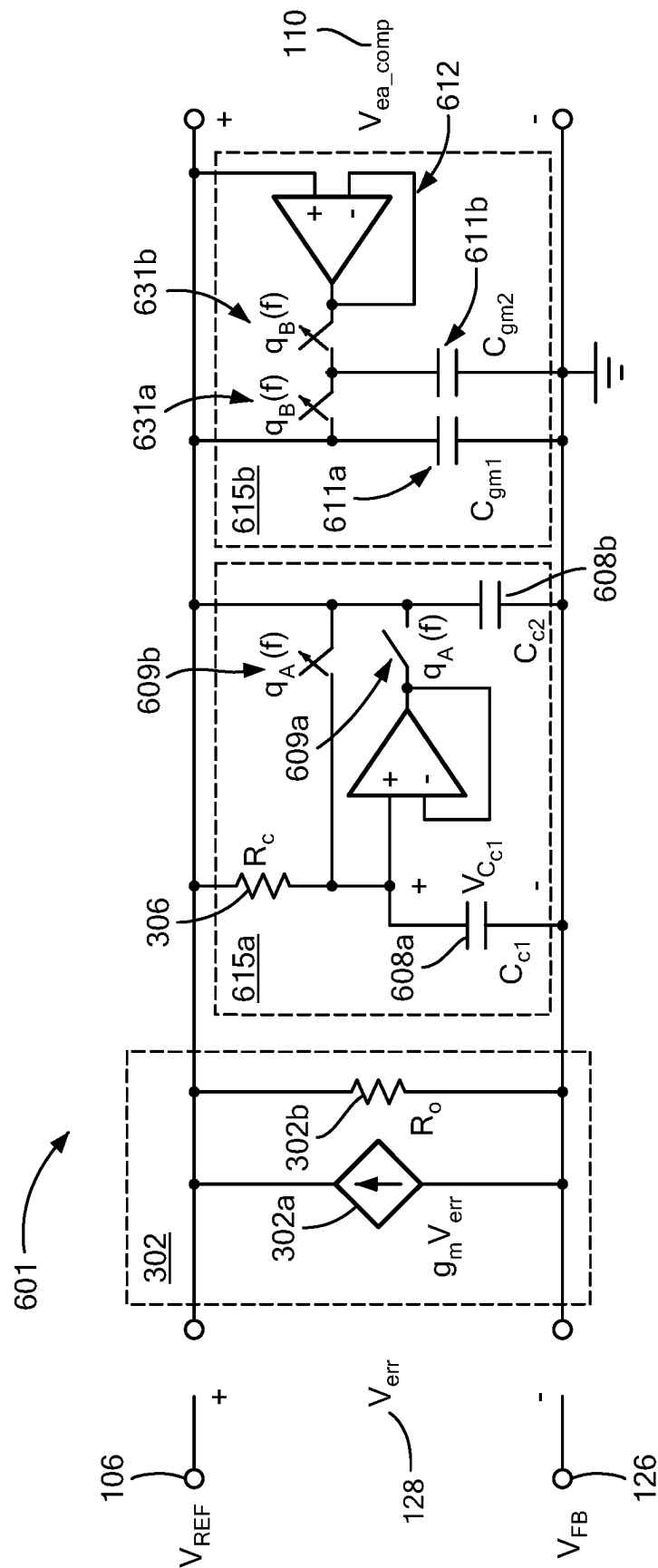
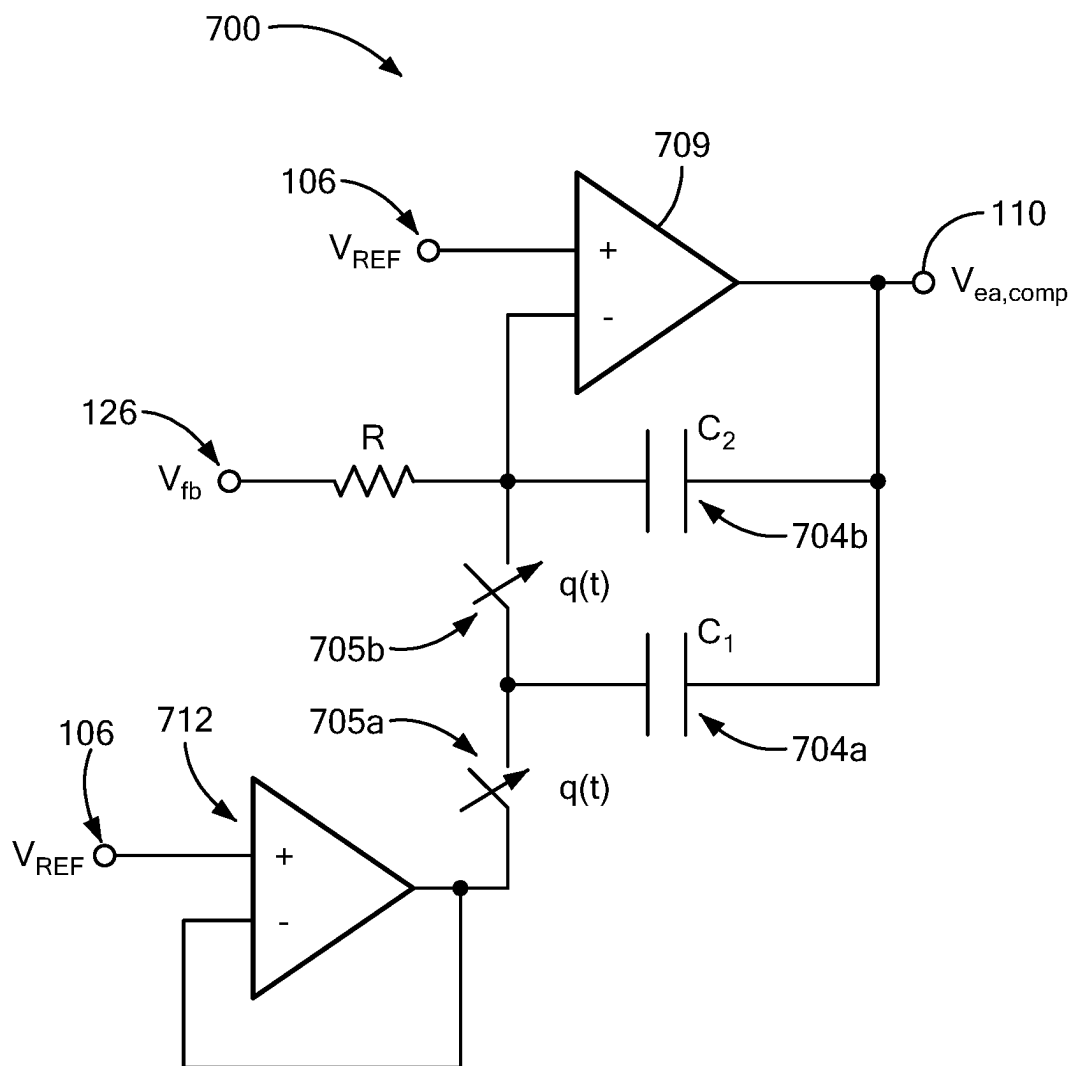


FIG. 6A



**FIG. 7**

1

VARIABLE CONTROLLER AND ASSOCIATED CONTROL METHODS

RELATED APPLICATION

This application claims the benefit of U.S. Provisional Application No. 63/235,927, filed Aug. 23, 2021, and entitled "VARIABLE CONTROLLER AND ASSOCIATED CONTROL METHODS," which is incorporated herein by reference in its entirety.

FIELD

This disclosure relates generally to a variable controller and associated control methods, and more particularly to a variable controller and associated control methods for a compensator.

BACKGROUND

As is known in the art, adjusting a control system to improve its performance might lead to unexpected behavior. For example, increasing a gain value of a control system may result in poor stability (or even instability) of the system being controlled. To make the control system behave as desired, it may be necessary to include a compensator network (also sometimes referred to as a "compensator system," a "compensator circuit," or more simply a "compensator") which compensates for deficient performance of an original control system. A compensator may thus be considered as component of a control system used to regulate other control system components. A compensator may perform such regulation, for example, by conditioning an input signal or an output signal of the system (e.g., conditioning an input or output signal to a control system).

As is also known, a power electronic converter is a circuit which converts one form or level of electrical power to another form or level (e.g., a dc-to-dc converter). Many power electronic converters must operate over both a range of operating characteristics as well as in multiple, different operating modes. For example, many power electronic converters must operate over a range of input and/or output signals, voltage conversion ratios, ranging load characteristics, a range of power levels and the like as well as in multiple, different operating modes (e.g., buck, buck+buck-boost, buck/boost, boost, peak current mode, valley current mode, low-power modes such as pulse frequency modulation, etc.).

The ability of a power converter to function over a range of operating characteristics and in multiple, different operating modes is often accomplished with a fixed controller or fixed compensator (i.e., a controller having a predefined architecture and controller structure such as a single-loop SISO control architecture where the controller is a fixed-order transfer function, a PID controller, or a PID controller plus a filter). However, because the dynamics of devices such as power converters vary considerably with operating point and operational mode, the achievable dynamic performance within a mode and/or over a range of power converter operating characteristics can be quite limited by use of a single fixed controller or compensator.

SUMMARY

In accordance with the concepts described herein, it has been recognized that it would be valuable to provide a compensator that yields different dynamics/control perfor-

2

mance for one or more of: (a) different operating modes (e.g., buck vs. boost vs. buck+buck-boost, or Pulse-Width Modulation vs. Pulse Frequency Modulation, etc.); and (b) different operating ranges or points (e.g., different input voltage, output voltage, reference voltage, conversion ratio or load regimes) such as different operating modes and/or different operating ranges or points of a power electronics circuit.

In accordance with the concepts described herein, it has been recognized that it would be further desirable to be able to change compensator characteristics and/or dynamics such as pole and/or zero locations (i.e., the frequencies for which the value of the denominator and numerator of a transfer function becomes zero, respectively and which determine the stability of the system represented by the transfer function) without introducing transients or "bumps" in operation at the transition point when compensator characteristics are changed.

Described is such a time/mode/operating point variable controller that is particularly well suited to integrated high-performance power electronic converter applications. It will also be appreciated that many other applications need operate across wide ranges and multiple modes and can likewise benefit from the techniques described herein.

In an aspect, a circuit includes a first capacitor configured to be charged to a voltage representing state information of a compensator, a second capacitor, a buffer circuit configured to charge the second capacitor to a voltage substantially equal to that of the first capacitor, and a switching network configured to transition between a first state and a second state. When the switching network is in the first state, the second capacitor is charged to the voltage across the first capacitor. When the switching network is in the second state, the buffer circuit is disconnected from the second capacitor and the first capacitor and the second capacitor are connected in parallel.

Implementations can include one or more of the following features.

In some implementations, when the switching network is in the first state, dynamics of the compensator are determined at least in part by the first capacitor, and when the switching network is in the second state, the dynamics of the compensator are determined at least in part by a total capacitance provided by the first capacitor and the second capacitor connected in parallel.

In some implementations, the capacitance provided based on the state of the switching network affects a pole location of the compensator.

In some implementations, the capacitance provided based on the state of the switching network affects a zero location of the compensator.

In some implementations, transitioning the switching network between the first and second states does not cause a transient impact on the voltage across the first capacitor.

In some implementations, transitioning the switching network between the first and second states does not cause a transient impact on a voltage provided at an output of the compensator.

In some implementations, transitioning the switching network between the first and second states does not disturb the voltage provided at the output of the compensator.

In some implementations, the switching network comprises a complementary switch pair configured to switch between the first state and the second state.

In some implementations, the circuit includes a third capacitor. The buffer circuit is configured to charge the third capacitor to a voltage substantially equal to the voltage

3

across the first capacitor. The switching network is configured to transition between at least two additional states to cause the third capacitor to switch between being charged to the voltage across the first capacitor, and being disconnected from the buffer circuit and connected in parallel with one or both of the first capacitor and the second capacitor.

In some implementations, the circuit includes a resistor connected to one or both of the first capacitor and the second capacitor.

In some implementations, the circuit is part of a power converter.

In some implementations, when the complementary switch pair is in the first state, a first switch is closed and a second switch is open. When the complementary switch pair is in the second state, the first switch is open and the second switch is closed.

In some implementations, the switching network is controlled by a control signal.

In some implementations, the control signal automatically causes switches of the switching network to switch based on one or both of an operating mode and an operating point of the circuit.

In some implementations, the operating mode includes buck, boost, or buck+boost mode.

In some implementations, the operating point includes one or more of an input voltage, an output voltage, a conversion ratio, and a load current of the circuit.

In some implementations, the buffer circuit includes an op-amp.

In some implementations, the op-amp is configured to charge the second capacitor to a voltage that is related to an output voltage and a reference voltage of the compensator.

In an aspect, a compensator circuit is configured to provide an output voltage. The compensator circuit includes a plurality of resistors and a switching network coupled to at least one of the plurality of resistors. The switching network is configured to transition between states to cause selected ones of the plurality of resistors to be connected in parallel with each other. The compensator circuit also includes a capacitor coupled to the selected ones of the plurality of resistors connected in parallel with each other.

Implementations can include one or more of the following features.

In some implementations, dynamics of the compensator circuit are determined at least in part by a total resistance provided by the selected ones of the plurality of resistors connected in parallel with each other.

In some implementations, the total resistance determined by the state of the switching network affects a zero location of the compensator circuit.

In some implementations, the compensator circuit is part of a power converter.

BRIEF DESCRIPTION OF THE DRAWINGS

The manner and process of making and using the disclosed embodiments may be appreciated by reference to the figures of the accompanying drawings. It should be appreciated that the components and structures illustrated in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principals of the concepts described herein. Like reference numerals designate corresponding parts throughout the different views. Furthermore, embodiments are illustrated by way of example and not limitation in the figures, in which:

FIG. 1 is a block diagram of an exemplary control system for a power converter;

4

FIG. 2 is a schematic diagram of an exemplary prior art compensator.

FIG. 3 is a schematic diagram of an exemplary compensator provided in accordance with the concepts described herein in which a resistance value of a resistor R_c can be dynamically changed;

FIG. 4 is a schematic diagram of an exemplary compensator having a capacitor C_{gm} which can be dynamically changed between a first capacitance value and a second capacitance value;

FIG. 5 is a schematic diagram of an exemplary compensator having a capacitor C_{gm} which can be dynamically changed among a plurality of capacitance values without disturbing a signal provided at an output of the compensator;

FIG. 6A is a schematic diagram of an exemplary compensator that includes a plurality of capacitive buffer circuits;

FIG. 6B is a schematic diagram of an exemplary compensator in which the values of capacitive buffer circuits can each be dynamically changed (or reprogrammed) among two (2) values; and

FIG. 7 is a schematic diagram of an exemplary compensator in which a capacitive value of a feedback capacitor C can be dynamically reprogrammed among two (2) values.

DETAILED DESCRIPTION

Referring now to FIG. 1, an exemplary system **100** includes a control system **101** (sometimes referred to herein as a controller) for a power stage **116** (e.g., a dc-dc power converter). The system **100** may be a power electronics system. The control system **101** includes a selectable voltage divider **104** having an input coupled to an input **102** of control system **101**. A bandgap reference voltage V_{egref} is provided to control system input **102** and coupled to an input of the selectable voltage divider **104**. Selectable voltage divider **104** receives the bandgap reference voltage V_{egref} and provides a target reference voltage V_{REF} at an output **106** thereof. As will become apparent from the description provided herein, target reference voltage V_{REF} for an output voltage V_{LN} **122** is set by bandgap voltage reference V_{egref} . In embodiments, bandgap voltage reference V_{egref} may have a value of about 1.25 V, which may be divided down by a programmable amount via selectable voltage divider **104**.

An output of selectable voltage divider **104** is coupled to a first input of a difference circuit **107** and a second input of difference circuit **107** is coupled to receive a feedback voltage V_{fb} **126**. An output of difference circuit **107** represents an error voltage V_{err} **128** between the output voltage V_{LN} **122** and a desired value of the output voltage. The output of difference circuit **107** is coupled to an input of a compensator **108** (sometimes referred to as an error amplifier), and an output of compensator **108** is coupled to a current control circuit **112**. An output of current control circuit **112** is coupled to the power stage **116** and power stage **116** provides an output voltage V_{LN} at an output **122** thereof. Power stage **116** may be provided, for example, as a power converter. It should, of course, be appreciated that although the broad concepts described herein are described in the context of a power electronics system, control systems (such as control system **101**) operating in accordance with the concepts described herein may be used in conjunction with other types of systems (i.e., systems other than power electronics systems). Such other systems can include servo-mechanisms (e.g., position or velocity control systems), tracking or targeting systems, and actuation systems, among many others.

5

In some implementations, the compensator **108** may be referred to as an error amplifier herein. In some implementations, the components of the control system **101** may be referred to cumulatively as a compensator, and an error amplifier can be a part of the overall compensator.

At least a portion of output voltage V_{LN} **122** is coupled to an input of a voltage divider circuit **124** and a feedback voltage V_{fb} **126** is provided at an output of voltage divider circuit **124**. Thus, feedback voltage V_{fb} **126** (which may be a divided down version of the output voltage V_{LN} **122** provided by voltage divider **124**) is provided to a second input of difference circuit **107** and compared with reference voltage V_{REF} to provide an error voltage V_{err} **128**.

Error voltage V_{err} **128** is coupled to an input of compensator **108**. Compensator **108** provides an output signal (here, an output voltage V_{ea_comp}) at an output **110** thereof. Compensator output signal is coupled through current control circuit **112** to an input of power stage **116** and is used to control power stage **116**. Such control system output signals may correspond, for example, to signals for commanding a duty ratio in a duty ratio-controlled converter (also known as voltage-mode control), commanding a peak commanded current for the inductor in a peak current-mode-controlled converter, or another control variable (e.g., duty ratio, valley current, average current, etc.).

In embodiments, an associated modulator (duty cycle generator, PWM unit, current-mode control block, etc.) may use such control signals (e.g., commands) to generate switching signals for the power stage **116**, as is known in the art.

An output signal from power stage **116** is coupled through a scaling and offset circuit **118** to an input of current control circuit **112** as a current signal i_{sense} . A ramp generator circuit **120** also provides a signal to current control circuit **112**. It should be appreciated that circuits **118**, **120** are auxiliary to the controller and thus in some embodiments may not be included or may not be considered part of control system **101**.

The current control circuit **112** provides a way to regulate inductor current in a power converter. The current control circuit **112** takes as inputs a desired reference for said inductor current V_{ea_comp} , provided at the output **110** of the compensator **108**, and a measurement or estimate of an inductor or other current (i_{sense}) in the power converter power stage **116**, generates as an output a switching signal $q(t)$ for said power converter power stage **116**. The current control circuit **112** can be useful for providing multi-loop control or full-state feedback control of a power converter for improved dynamic performance. Scaling and offset circuit **118** allows a measured or estimated value of the inductor or other current (i_L) in power stage **116** to be converted to a form and range that is useful for the current control circuit **112** (i.e., i_{sense}). Ramp generator **120** can provide a timing (clocking) reference and ramp to the current control circuit **112** allow the current control circuit **112** to implement slope compensation.

Characteristics of the compensator **108** can be selected based upon the dynamics of the “plant.” In some implementations, the plant comprises at least the power stage **116**. The characteristics can be selected to result in stable control of the output of the system **100**, preferably with fast transient response and good stability margin. The plant dynamics (i.e., the open-loop dynamics of power stage **116**), and hence the overall system dynamics (i.e., the closed loop system dynamics of the plant—i.e., power stage **116**—modified by the compensator **108**), can depend upon converter operating mode (buck, boost, PFM, PWM, etc.) as well as operating

6

point (output voltage, load, conversion ratio, etc.) making the design of the compensator **108** challenging.

The compensator **108** of FIG. **1** may take the form of or may include one or more features of any one or more of the compensators described herein with respect to FIGS. **3-7**.

Referring now to FIG. **2**, FIG. **2** shows a circuit diagram of a prior art compensator **200**. Such a circuit might be implemented as an integrated circuit (IC) such as part of a power converter IC or as a controller IC.

Compensator **200** comprise an amplifier **202** which is here represented as a model having a gain element **202a** and an output resistance R_o **202b**. In embodiments, amplifier **202** may, for example, be provided as a transconductance amplifier (e.g., an operational transconductance amplifier) or a voltage amplifier. A plurality of resistive and/or capacitive signal paths (or “legs”) **205**, **209** are coupled at an output of amplifier **202**, for example, in the form of an output voltage V_{ea_comp} . In this example, signal path **205** comprises series coupled resistor R_c **206** and a capacitor C_c **208** and signal path **209** comprises a capacitor C_{gm} **210**. In embodiments, compensator **200** may be provided as an integrated circuit.

In some implementations, amplifier **202** might be realized as an operational amplifier. In some implementations, the value of gain (“transconductance”) g_m might be made programmable over a range in order to allow some degree of programmability/dynamic tuning of the system dynamics.

An error voltage V_{err} **228** is provided to an input of compensator **200** and coupled to amplifier **202**.

If one assumed that amplifier output resistance R_o **202b** were infinite, a small-signal compensator transfer function $G_C(s)$ from $V_{err}=V_{REF}-V_{fb}$ to the compensator output **110** V_{ea_comp} could be expressed as:

$$G_C(s) = \frac{V_{ea_comp}}{V_{err}} = \frac{g_m}{s(C_{gm} + C_c)} \cdot \frac{1 + sR_c C_c}{1 + sR_c \frac{C_{gm} C_c}{C_{gm} + C_c}}. \quad (\text{Eq. 1})$$

In practical circuits, however, the output resistance R_o **202b** is finite. However, if the amplifier output resistance is sufficiently large (e.g., such that the error between (Eq. 1) and (Eq. 2) is unimportant), this expression is useful for approximately locating the compensator zero and second pole. Including output resistor R_o **202b**, the following compensator transfer function results:

$$G_C(s) = \frac{V_{ea_comp}}{V_{err}} = g_m R_o \cdot \frac{1 + sR_c C_c}{s^2 R_o C_{gm} R_c C_c + s(R_o C_{gm} + R_o C_c + R_c C_c) + 1}. \quad (\text{Eq. 2})$$

In this exemplary compensator **200**, the compensator DC gain is set by the product of amplifier gain g_m and output resistance (i.e., the compensator DC gain is set by the product of $g_m R_o$). The value of resistor R_c **206** and capacitor C_c **208** set the zero of the transfer function, while the pole locations are set by multiple component parameters as shown in Eq. 2. One can adjust compensator dynamics (i.e., the manner in which the compensator responds—in an electrical circuit sense—to inputs provided thereto) by introducing programmability of one or more of these elements (e.g., by using a switchable resistor bank to dynamically adjust (or change) the resistance value of resistor R_c **206** to modify the compensator zero, by dynamically programming the gain value g_m to modify the compensator gain, etc.).

Such dynamic adjustments/changes/programming can be based on system operating condition (e.g., mode, operating voltages or power levels, etc.). It can be desirable to have different values of compensator components to modify how the circuit will react dynamically under different system operating conditions, and such dynamic adjustability can allow such.

The capacitor C_{gm} 210 can be a combination of device output capacitance and physical capacitance used in setting characteristics of the transfer function $G_C(s)$ as per (Eq. 2). The resistor R_c 206 and the capacitor C_c 208 are circuit elements used to set the transfer function as per (Eq. 2). Using the components as illustrated allows the ability to set different characteristics of the transfer function $G_C(s)$.

Referring now to FIG. 3, in this embodiment, an exemplary compensator 300 includes a programmable resistive element 306 (i.e., a resistive element capable of being adjusted or changed) comprising a plurality (or bank) of parallel coupled resistive legs 305a-405N. At least one of the resistive legs (illustrated as a single resistive leg 305a in FIG. 3) comprises a resistor (here illustrated as resistor 306a) having a resistance value R_{c0} . The compensator 300 also includes a switching network that can include a plurality of switch elements, or switches S_1 - S_N 307b-307N.

Programmable resistive element 306 further comprises at least one resistive leg comprising both a resistive element and a switch element. In the example embodiment of FIG. 3, resistive legs 305b-305N each comprise both a resistive element and a switching element. Taking resistive leg 306b as illustrative of resistive legs 305c-305N, resistive leg 306b comprises a resistor 306b having a resistance value R_{c1} coupled to a switch S_1 307b. Similarly, resistive legs 305c-305N comprise respective ones of resistors 306c-306N having respective resistance values R_{c2} - R_{cN} coupled to respective ones of switches S_2 - S_N 307c-307N.

When switches $S_1, \dots, S_N$ 307b-N are closed or on (i.e., the switch is biased or otherwise set into a state or position such that a low impedance signal path (and ideally a short circuit impedance signal path) exists between a pair of terminals thereof) a conductive path having a resistance less than an open circuit resistance exists between nodes 313a, 313b. For example, with switch S_1 closed or on and all other switches $S_2, \dots, S_N$ open or off, a signal path comprising resistor 306b and switch 307b having a resistance substantially equal to R_{c1} exists between nodes 313a, 313b. Thus, by selecting which of switches $S_1, \dots, S_N$ 307b-N are turned on, resistive element 306 is adjustable (or programmable or selectable) in steps between a first (or minimum) resistance value and a second (or maximum) resistance value.

In the example embodiment of FIG. 3, the maximum resistance value is set by the resistance value R_{c0} of resistor 306a and the minimum resistance value is set by the equivalent resistance provided by resistors 306a-306N (i.e., the equivalent resistance provided by the parallel combination of resistors $R_{c0} || R_{c1} || \dots || R_{cN}$). Thus, by selecting which of switches $S_1, \dots, S_N$ 307b-N are turned on (i.e., biased or otherwise set into their ON states) resistive element 306 is provided having one of a plurality of different resistance characteristics.

The plurality of switches S_1 - S_N 307b-307N can thus operate (or be controlled) to cause one or more of the plurality of second resistors R_{c1} - R_{cN} 306b-306N to either be electrically disconnected from the first resistor R_{c0} 306a or coupled in parallel with the first resistor R_{c0} 306a. A total resistance provided by the first resistor R_{c0} 306a and coupled ones of the plurality of second resistors R_{c1} - R_{cN} 306b-306N can cause one or more dynamics of the compensator 300 to

be adjusted, such as one or more zero locations of the compensator 300, as described herein.

As noted above, resistive element 306 has a first terminal coupled to a node 313a at which a compensator output 110 (e.g., output voltage V_{ea_comp}) is provided. For example, the voltage at node 313a with reference to ground may correspond to the compensator output 110 (e.g., the output voltage V_{ea_comp}). A second terminal of resistor 306a is coupled to a node 313b. A first terminal of a capacitor 308 having a capacitance of C_c is coupled to a node 313b and a second terminal of capacitor 308 coupled to a reference potential, here ground.

It should, of course, be appreciated that while FIG. 3 illustrates one example embodiment of a dynamically programmable element (i.e., a dynamically programmable resistance element 306) disposed to influence (e.g., change or shape) the dynamics of a compensator, other embodiments which implement the concept of dynamic programmability are also possible. For example, an implementation comprising a programmable resistance element may comprise a first signal path having a set of individual resistors coupled in series, a second signal path coupled in parallel to the first signal path and or more control switches (e.g., one or more of switches 307b-307N) disposed at various points between the first and second signal paths and operable to selectively include or bypass one, some or all of the resistors in the first signal such that only selected ones of the resistors in the first signal path contribute to the overall resistance between first and second terminals of the programmable resistance element and thus can be used to adjust one or more dynamics of the compensator. For example, the one or more dynamics that can be adjusted can include one or more zero locations of the compensator.

In another example embodiment, the plurality of second resistors R_{c1-N} 306b-N and the plurality of switches S_{1-N} 307b-N can be arranged such that the switches 307b-N operate to couple selected ones of resistors R_{c1-N} 306b-N in series with the first resistor R_{c0} 306a. A total resistance provided by the first resistor R_{c0} 306a and the plurality of second resistors R_{c1-N} 306b-N can cause one or more dynamics of the compensator 300 to be adjusted. For example, the one or more dynamics that can be adjusted can include one or more zero locations.

The above-described programmability can be such that a dynamic (transient) impact on the system does not result during operation. In the embodiment of FIG. 3, compensator 300 comprises a programmable resistance element and the compensator 300 state is held by the voltage on capacitors C_{gm} 311 and C_c 308, with capacitor C_{gm} 311 being particularly central to determining the compensator pole locations and resulting system dynamics. Stated differently, in the system of FIG. 3, the compensator output 110 V_{ea_comp} is substantially, directly determined by the voltage on capacitor C_{gm} 311. In this way, the voltage that the capacitor C_{gm} 311 is charged to can represent state information (e.g., the compensator dynamics described herein such as gain, pole locations, zero locations, etc.) for the compensator 300. Any operation (e.g., compensator programming change) that disturbs this voltage causes a variation (e.g., a "bump") in the compensator output 110 V_{ea_comp} . Such variations in compensator output signals (e.g., output voltage V_{ea_comp}) are undesirable for control.

Thus, in accordance with a further aspect of the concepts described herein, it has been recognized that it would be desirable to provide a system that allows dynamic reprogrammability/variability of compensator capacitors (such as capacitor C_{gm} 311) to enable rapid modification of control

(and associated dynamics) for different modes and operating ranges without introducing control bumps in the compensator output voltage V_{ea_comp} .

In accordance with a still further aspect of the concepts described herein, it has also been recognized that the above goal may be achieved by introducing one or more additional compensation capacitors that can be switched in and out (i.e., electrically coupled to and electrically decoupled from) a compensator network thus making the compensation capacitor dynamically programmable.

As will be described in detail below, to prevent undesirable transients when a compensation capacitor is switched into the compensator, a buffer may be utilized that keeps compensation capacitors that are not presently active in the compensator pre-charged at the value of the compensator capacitor voltage in use (i.e., tracking the active compensator capacitor's voltage value.) An exemplary circuit for realizing a programmable value for a capacitor (e.g., such as capacitor 311 in FIG. 3 having capacitance value C_{gm}) is shown in FIG. 4.

Referring now to FIG. 4, shown is an exemplary compensator 400 having an effective capacitance which can be dynamically programmed (i.e., changed or otherwise adjusted during operation) between at least a first capacitance value and a second, different capacitance value. In the example of FIG. 4, the capacitance of compensator 400 can be dynamically programmed between a first capacitance value corresponding to C_{gm1} and a second capacitance value corresponding to $C_{gm1} + C_{gm2}$. Significantly, the effective capacitance value of compensator 400 (generally referred to as C_{gm}) can be programmed without disturbing the compensator output signal 110 (e.g., without causing a transient impact on the compensator output voltage V_{ea_comp}).

In this example embodiment, compensator 400 includes a first capacitor 411a and a capacitive buffer circuit 415. In this example, the capacitive buffer circuit 415 includes a second capacitor 411b. First capacitor 411a is provided having a capacitance corresponding to C_{gm1} and the second capacitor 411b is provided having a capacitance corresponding to C_{gm2} . Capacitive buffer circuit 415 also comprises a switching network 416 coupled to the second capacitor 411b. Switching network 416 has two states: a first state for charging the second capacitor 411b to a desired value, and another state for placing capacitor 411b in parallel with capacitor 411a. It should be appreciated that the functionality of switching network 416 may be achieved with a variety of different circuit implementations. For example, as illustrated in the example embodiment of FIG. 4, such a switching network 416 may be implemented via a single complementary switch pair such as with switches 431a, 431b. In FIG. 4, switches 431a, 431b are configured to switch in a complementary fashion (i.e., when switch 431a is biased into its ON state, switch 431b is biased into its OFF state and vice-versa) and thus switches 431a, 431b are sometimes referred to as a "complementary switch pair" or as "complementary switches." After reading the disclosure provided herein, those of ordinary skill in the art will appreciate how best to implement the functionality provided by a switching network for use in a particular application.

Complementary switch pair 431a, 431b are controlled by a switching signal $q(t)$ (also sometimes referred to as a "switch control signal," a "programming signal" or more simply a "control signal"). The complementary switch pair 431a, 431b are coupled to a first capacitor 411a having a capacitance of C_{gm1} .

In one example embodiment, buffer circuit 415 is configured to receive a switch control signal $q(t)$ such that when

switch control signal $q(t)$ has a first value (e.g., $q(t)$ is low such as a logical 0), switch 431a is closed (or placed in its ON state) and switch 431b is open (or placed in its OFF state).

In this circuit configuration (i.e., control signal $q(t)$ has the first value thereby placing switch 431a in closed or ON state and switch 431b in an open or OFF state), capacitor 411a is coupled to a buffer 412 that ensures its voltage tracks that of the output voltage 110, and includes switch circuitry which effectively electrically disconnects (or electrically removes) capacitor 411a from the compensator circuit 400 (e.g., such that the capacitor 411a having capacitance value C_{gm1} does not influence the dynamics of the compensator 400). In embodiments, the buffer 412 may comprise an op-amp follower circuit or more simply "a follower" that maintains the voltage on the first capacitor 411a to be substantially the same as the voltage on a second capacitor 411b, which is also substantially the compensator output voltage V_{ea_comp} 110.

A first input of buffer 412 (here a positive input) can be coupled to a node 413 (e.g., a positive node of the compensator output 110 with reference to ground) and be configured to charge the first capacitor 411a to substantially the same voltage provided at the node 413 (e.g., the compensator output voltage V_{ea_comp}).

Continuing with the above example, when control signal $q(t)$ has a second value different from the first value (e.g., control signal $q(t)$ is high or a logical 1), the first switch 431a is open (or biased or otherwise set into its open or off state) and the second switch 431b is closed (or biased or otherwise set into its closed or on state) such that the first capacitor 411a is electrically disconnected from the buffer 412 and is coupled in parallel with capacitor 411b such that the compensator has an effective capacitance value of $C_{gm} = C_{gm1} + C_{gm2}$.

As noted above, the complementary switches 431a, 431b are configured to transition between a first state and a second state. When the complementary switches 431a, 431b are in the first state, the first capacitor C_{gm1} 411a can be charged by buffer 412 to the voltage provided at the node 413 (e.g., the compensator output voltage V_{ea_comp}) without capacitor C_{gm1} 411a affecting how the compensator behaves. When the complementary switches 431a, 431b are in the second state, the first capacitor C_{gm1} 411a and the second capacitor C_{gm2} 411b are coupled in parallel to the node, operating together as a single capacitor.

In embodiments, when the switch control signal $q(t)$ changes from a first value to a second different value (e.g., from a logical 0 to a logical 1), the effective compensator capacitor C_{gm} 411 changes to $C_{gm1} + C_{gm2}$ from C_{gm1} . However, because capacitor C_{gm2} 411b may be appropriately precharged by the buffer 412, the compensator output voltage V_{ea_comp} is not disturbed (e.g., no transient impact is caused on the voltage across C_{gm1} 411a and/or C_{gm2} 411b). Likewise, when switch control signal $q(t)$ changes from the second, different value to the first value (e.g., from a logical 1 to a logical 0), the effective compensator capacitor 411 changes to C_{gm1} from $C_{gm1} + C_{gm2}$ without disturbing the compensator output. Thus, depending on the state of the complementary switch pair and the total effective compensator capacitance provided by capacitor 411 determined by switch control signal $q(t)$, different pole locations for the compensator may be provided. In this way, the total effective compensator capacitance provided by capacitor 411 can cause one or more dynamics of the compensator 400 to be adjusted.

11

It will be appreciated that with additional capacitors and complementary switch pairs (with corresponding control signals) the range of programmability of a compensation capacitor may be increased.

FIG. 5, for example, shows a circuit diagram of an exemplary compensator 500 comprising a plurality of compensation capacitors. In this example embodiment, compensator 500 comprises three (3) compensation capacitors. After reading the disclosure provided herein, those of ordinary skill in the art will appreciate that compensator 500 may comprise fewer or more than three compensation capacitors. In this example, compensator 500 comprises a first compensation capacitor 511a having a capacitance value of C_{gm1} , a second compensation capacitor 511b having a capacitance value of C_{gm2} , and a third compensation capacitor 511c having a capacitance value of C_{gm3} .

With the plurality of compensation capacitors in the example embodiment of FIG. 5, the effective capacitance value of compensator 500 can be dynamically reprogrammed (i.e. adjusted or changed) among 4 values: C_{gm1} , $(C_{gm1}+C_{gm2})$, $(C_{gm1}+C_{gm3})$, and $(C_{gm1}+C_{gm2}+C_{gm3})$ without disturbing (i.e., without disrupting or otherwise substantially changing) the voltage V_{ea_comp} at the compensator output 110.

In this example embodiment, programming (or adjustment or changing) of the compensator capacitance is done via two control signals $q_2(t)$ and $q_3(t)$ (i.e., switch control signals) that control a switching network that can include complementary switch pairs 531a, b (generally denoted 531), 533a, b (generally denoted 533). Each of the control signals $q_2(t)$ and $q_3(t)$ controls one of the complementary switch pairs 531, 533, and the control signals $q_2(t)$ and $q_3(t)$ cause corresponding capacitors 511b and 511c to be switched between participating in the compensation and being precharged by a buffer 512. The complementary switch pairs 531, 533, the buffer 512, and the capacitors C_{gm2} - C_{gm3} 511b-c are part of a capacitive buffer circuit 515 of the compensator 500. In this example, control signal $q_2(t)$ controls complementary switch pair 531a, 531b and control signal $q_3(t)$ controls complementary switch pair 533a, 533b.

The first complementary switch pair 531a, 531b is coupled to the capacitor C_{gm2} 511b and second complementary switch pair 533a, 533b is coupled to the capacitor C_{gm3} 511c (sometimes referred to herein as a third capacitor, e.g., relative to the first and second capacitors of FIG. 4). The first complementary switch pair 531a, 531b and the capacitor 511b are configured to operate similarly to the first capacitor 411a and the complementary switch pair 431 of FIG. 4.

The second complementary switch pair 533 is configured to transition between a first state and a second state. When the second complementary switch pair 533 is in the first state, a first switch 533a is closed (i.e., switch 533a is in its ON state) and a second switch 533b is open (i.e., switch 533b is in its OFF state) such that the third capacitor 511c can be charged to a voltage provided at a node 513 to which a positive input of the buffer 512 is coupled. In the first state, the third capacitor C_{gm3} 511c can be coupled to an output of the buffer 512 to charge to the voltage V_{ea_comp} provided at the compensator output 110. The compensator output voltage V_{ea_comp} is provided at the node 513 relative to ground. When the second complementary switch pair 533 is in the second state, the first switch 533a is open and the second switch 533b is closed such that the third capacitor C_{gm3} 511c can be coupled to the node 513 in parallel with C_{gm2} 511b and/or C_{gm1} 511a (e.g., which can correspond to the first and second capacitors of FIG. 4).

12

Generally, augmenting a base capacitor (e.g., capacitor 511a) with N complementary switch pairs and associated programming capacitors will allow selection among up to 2^N programmable values of compensation capacitance.

It will be further appreciated that compensator designs can be implemented having multiple programmable compensation capacitors.

Referring now to FIG. 6A in which like elements of FIG. 4 are provided having like reference designations, shown is a circuit diagram of an exemplary compensator 600 that may be generally similar to the compensators described in the preceding figures. The circuit diagram expresses the concept that the compensators described herein may include one or more capacitive buffer circuits 615a, 615b, . . . 615n. Each of the capacitive buffer circuits 615 can include at least one or more capacitors, one or more resistors, one or more buffers, and a switching network that can include one or more switches that are configured to couple the elements in a way that causes one or more of the dynamics of the circuit to be adjusted without causing a transient impact on the output voltage V_{ea_comp} at the compensator output 110.

Referring now to FIG. 6B in which like elements of FIG. 6A are provided having like reference designations, a circuit diagram of an exemplary compensator 601 in which the values of compensator capacitors C_{gm} 611 and C_c 608 are each dynamically reprogrammable among two values. Compensator 601 comprises first and second capacitive buffer circuits 615a, 615b. The right-half of the circuit diagram comprises second capacitive buffer circuit 615b which may be the same as or similar to capacitive buffer circuit 615b shown in FIG. 6A). Capacitive buffer circuit 615b may operate in a manner which is the same as or similar to the capacitive buffer circuit described above in conjunction with FIG. 4, with a buffer 612 configured to charge a capacitor 611b and a switching network including a complementary switch pair 631 configured to switch between two states to cause the capacitor 611b to switch between being charged by the buffer 612 and being coupled to another capacitor 611a in parallel.

The left-half of the circuit diagram, corresponding to capacitive buffer circuit 615a, which may be the same as or similar to capacitive buffer circuit 615b shown in FIG. 6A, employs a concept similar to the ones described with respect to the preceding figures. In particular, the compensator 600 includes a first capacitor C_{c1} 608a that is coupled to a positive input of a buffer 607, a second capacitor C_{c2} 608b coupled to an output of the buffer 607, and a complementary switch pair 609 configured to alternate between a first state and a second state. In the first state, a first switch 609a is closed and a second switch 609b is opened such that the second capacitor C_{c2} 608b charges to a voltage equal to a voltage across the first capacitor C_{c1} 608a. In the second state, the first switch 609a is open and the second switch 609b is closed such that the second capacitor C_{c2} 608b is coupled in parallel with the first capacitor C_{c1} 608a. When coupled in parallel, the combined capacitance of the first capacitor C_{c1} 608a and the second capacitor C_{c2} 608b is coupled in series with the resistor R_c 406.

A control signal q_A can be used for programming the capacitance used for capacitors C_{c1} 608a and C_{c2} 608b, and a control signal q_B can be used for programming the capacitance used for capacitors C_{gm1} 611a and C_{gm2} 611b. One could likewise have circuits in which each programmable in multiple steps as illustrated in the preceding figures.

The capacitor value(s) used in the compensator can be automatically adjusted based on operating mode (e.g., buck vs. boost, etc.), operating point (e.g., input voltage, output

voltage, conversion ratio, load current value, etc.) or other range associated with converter operation, or based on associated reference or target values. Alternatively or additionally, the values may be made software or firmware programmable, e.g. using registers that control the control signals $q(t)$.

While the above examples focus on various types of compensators and implementations that are easily realized for IC power converter designs, it will be appreciated that a variety of compensator types and implementations can benefit from the programmability described herein. For example, one can realize a variety of compensator types with programmable capacitor values (e.g., integral, lead, lag, lead-lag, PI, PID, PD, type I, type II, type III, etc.) Moreover, other circuit implementations for compensators can be used, such as compensator designs using an op amp as an error amplifier.

FIG. 7 shows a circuit diagram of an exemplary compensator **700** based on an op-amp **709** in which a net feedback capacitance C is provided by various combinations of capacitors **704a** **704b** (generally denoted **704**). The net capacitance value C can be dynamically selected to correspond to one of two values (C_1 and $C_1 + C_2$) via control signal $q(t)$. The compensator **700** includes a switching network that can include a complementary switch pair **705a**, **705b** configured to switch between a first state and a second state. In the first state, first switch **705a** is closed and second switch **705b** is open such that capacitor C_2 **704b** is charged to a reference voltage V_{REF} **706** by a buffer **712**. In this way, capacitor C_1 **704a** is pre-charged to the voltage it would have in the compensator **700** when it participates in the compensator **700** (e.g., when it is switched into the compensator **700** in such a way that it can have an influence on dynamics of the compensator **700**). This voltage can have the same value as the “active” capacitor it will be placed in parallel with (e.g., capacitor C_2 **704b**). In some implementations, the capacitor C_1 **704a** is pre-charged to a voltage related to the output voltage V_{ea_comp} **110** and the feedback voltage V_{fb} **126**. In the second state, first switch **705a** is open and second switch **705b** is closed such that capacitor C_1 **704a** is coupled in parallel with capacitor C_2 **704b**. In this way, the dynamics of the compensator **700** can be adjusted without causing a transient impact on the compensator output voltage V_{ea_comp} **110**.

It will be appreciated that while this is a type I or integral controller, other controller types (including, but not limited to on/off controllers, proportional controllers derivative controllers) with one or more programmable circuit elements (including, but not limited to one or more programmable capacitors and/or one or more programmable resistors) may be realized by adding or modifying the passive component networks around the op-amp **809** accordingly.

Moreover, it is also possible to use the proposed technique to adjust other compensator aspects. For example, by appropriately switching capacitor values and/or resistor values or gains, one can dynamically adjust the slope associated with slope compensation in a current-mode control circuit. This can be done as described for varying operating points, varying modes (e.g., buck vs. boost), or even within a switching cycle when transitioning from one mode or sub-mode to another (e.g., in moving from buck to extended buck+buck-boost mode). Such a multi-slope compensating ramp could be valuable in improving not only low-frequency dynamics over multiple modes and wide operating ranges but also ripple dynamics (i.e., as associated with “ripple instability” in current-mode-controlled converters.) One can likewise dynamically adjust PWM ramp slope in

this manner, either on a long-term basis or within a switching cycle. These functions may be accomplished because the associated waveforms are generated by charging a capacitor via a current source or resistor, and the capacitor and/or resistor and/or current source can be dynamically reprogrammed as described herein.

Although reference is sometimes made herein to use of the described compensators with power electronics and/or as part of a controller, it should be appreciated that compensators provided in accordance with the concepts described herein may find use in a wide variety of applications including any applications which operate with a control system. It is also recognized that compensators may be designed using electrical, mechanical, pneumatic or any other components. It should thus be appreciated that elements (including circuit elements) other than those specifically disclosed herein may be substituted where appropriate to suit the needs of a particular application, and that a person having ordinary skill in the art would understand how to select such elements and incorporate them into embodiments operating in accordance with the concepts described herein without deviating from the scope of the concepts described herein.

Various embodiments of the concepts, systems, devices, structures and techniques sought to be protected are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the concepts, systems, devices, structures and techniques described herein. It is noted that various connections are set forth between elements in the above description and in the drawings. These connections, unless specified otherwise, can be direct or indirect, and the described concepts, systems, devices, structures and techniques are not intended to be limiting in this respect. Accordingly, reference to coupled elements or a coupling of elements can refer to either a direct or an indirect coupling.

As an example of an indirect relationship, references in the present description to coupling element “A” to element “B” include situations in which one or more intermediate elements (e.g., element “C”) is between element “A” and element “B” as long as the relevant characteristics and functionalities of element “A” and element “B” are not substantially changed by the intermediate element(s). The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

Additionally, the term “exemplary” is used herein to mean “serving as an example, instance, or illustration. Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “one or more” and “one or more” are understood to include any integer number greater than or equal to one, i.e., one, two, three, four, etc. The terms “a plurality” are understood to include any integer number greater than or equal to two, i.e., two, three, four, five, etc. The term “connection” can include an indirect “connection” and a direct “connection.”

References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that

15

the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms “overlying,” “atop,” “on top,” “positioned on” or “positioned atop” mean that a first element, such as a first structure, is present on a second element, such as a second structure, where intervening elements such as an interface structure can be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary elements.

Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

The terms “approximately” and “about” may be used to mean within $\pm 20\%$ of a target value in some embodiments, within $\pm 10\%$ of a target value in some embodiments, within $\pm 5\%$ of a target value in some embodiments, and yet within $\pm 2\%$ of a target value in some embodiments. The terms “approximately” and “about” may include the target value. The term “substantially equal” may be used to refer to values that are within $\pm 20\%$ of one another in some embodiments, within $\pm 10\%$ of one another in some embodiments, within $\pm 5\%$ of one another in some embodiments, and yet within $\pm 2\%$ of one another in some embodiments.

The term “substantially” may be used to refer to values that are within $\pm 20\%$ of a comparative measure in some embodiments, within $\pm 10\%$ in some embodiments, within $\pm 5\%$ in some embodiments, and yet within $\pm 2\%$ in some embodiments. For example, a first direction that is “substantially” perpendicular to a second direction may refer to a first direction that is within $\pm 20\%$ of making a 90° angle with the second direction in some embodiments, within $\pm 10\%$ of making a 90° angle with the second direction in some embodiments, within $\pm 5\%$ of making a 90° angle with the second direction in some embodiments, and yet within $\pm 2\%$ of making a 90° angle with the second direction in some embodiments.

It is to be understood that the disclosed subject matter is not limited in its application to the details of construction and to the arrangements of the components set forth in the description or illustrated in the drawings. The disclosed subject matter is capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the

16

several purposes of the disclosed subject matter. Therefore, the claims should be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the disclosed subject matter.

Although the disclosed subject matter has been described and illustrated in the foregoing exemplary embodiments, it is understood that the present disclosure has been made only by way of example, and that numerous changes in the details of implementation of the disclosed subject matter may be made without departing from the spirit and scope of the disclosed subject matter.

What is claimed is:

1. A circuit comprising:

a first capacitor configured to be charged to a voltage representing state information of a compensator;

a second capacitor;

a buffer; and

a switching network configured to transition between a first state and a second state, wherein when the switching network is in the first state, the second capacitor is connected to the output of the buffer and charged to the voltage across the first capacitor, and wherein when the switching network is in the second state, the buffer is disconnected from the second capacitor and the first capacitor and the second capacitor are connected in parallel.

2. The circuit of claim 1, wherein when the switching network is in the first state, dynamics of the compensator are determined at least in part by the first capacitor, and when the switching network is in the second state, the dynamics of the compensator are determined at least in part by a total capacitance provided by the first capacitor and the second capacitor connected in parallel.

3. The circuit of claim 2, wherein switching between the first state and the second state of the switching network affects a capacitance of the circuit, which affects a pole location of the compensator.

4. The circuit of claim 2, wherein switching between the first state and the second state of the switching network affects a capacitance of the circuit, which affects a zero location of the compensator.

5. The circuit of claim 1, wherein transitioning the switching network between the first and second states does not cause a transient impact on the voltage across the first capacitor.

6. The circuit of claim 1, wherein transitioning the switching network between the first and second states does not cause a transient impact on a voltage provided at an output of the compensator.

7. The circuit of claim 6, wherein transitioning the switching network between the first and second states does not disturb the voltage provided at the output of the compensator.

8. The circuit of claim 1, wherein the switching network comprises a complementary switch pair configured to switch between the first state and the second state.

9. The circuit of claim 8, wherein when the complementary switch pair is in the first state, a first switch is closed and a second switch is open, and when the complementary switch pair is in the second state, the first switch is open and the second switch is closed.

10. The circuit of claim 1, further comprising:

a third capacitor, wherein the buffer is configured to charge the third capacitor to a voltage substantially equal to the voltage across the first capacitor, and wherein the switching network is configured to transition between at least two additional states to cause the third

17

capacitor to switch between being charged to the voltage across the first capacitor, and being disconnected from the buffer and connected in parallel with one or both of the first capacitor and the second capacitor.

11. The circuit of claim 1, further comprising a resistor
connected to one or both of the first capacitor and the second
capacitor.

12. The circuit of claim 1, wherein the circuit is part of a
power converter.

13. The circuit of claim 1, wherein the switching network
is controlled by a control signal.

14. The circuit of claim 13, wherein the compensator is
part of a system having a power converter, wherein the
control signal automatically causes switches of the switch-
ing network to switch based on one or both of an operating
mode and an operating point of the power converter.

15. The circuit of claim 13, wherein the control signal
automatically causes switches of the switching network to
switch based on an operating mode of a power converter,
wherein the operating mode comprises at least one of a buck,
boost, or buck+buck-boost mode.

16. The circuit of claim 13, wherein the control signal
automatically causes switches of the switching network to
switch based on an operating point of a power converter,
wherein the operating point comprises one or more of an
input voltage, an output voltage, a conversion ratio, or a load
current of the power converter.

17. The circuit of claim 1, wherein the buffer comprises
an op-amp.

18. The circuit of claim 17, wherein the op-amp is
configured to charge the second capacitor to a voltage that
is related to an output voltage and a reference voltage of the
compensator.

18

19. A compensator circuit configured to provide an output
voltage, the compensator circuit comprising:

a first capacitor configured to be charged to a voltage
representing state information of the compensator cir-
cuit;

a second capacitor;

a buffer; and

a first switching network configured to transition between
a first state and a second state, wherein when the first
switching network is in the first state, the second
capacitor is connected to the output of the buffer and
charged to the voltage across the first capacitor, and
wherein when the first switching network is in the
second state, the buffer circuit is disconnected from the
second capacitor and the first capacitor and the second
capacitor are connected in parallel;

a plurality of resistors;

a second switching network coupled to at least one of the
plurality of resistors, wherein the second switching
network is configured to transition between states to
cause selected ones of the plurality of resistors to be
connected in parallel with each other; and

a third capacitor coupled to the selected ones of the
plurality of resistors connected in parallel with each
other.

20. The compensator circuit of claim 19, wherein dynam-
ics of the compensator circuit are determined at least in part
by a total resistance provided by the selected ones of the
plurality of resistors connected in parallel with each other.

21. The compensator circuit of claim 20, wherein the total
resistance affects a zero location of the compensator circuit.

22. The compensator circuit of claim 19, wherein the
compensator circuit is part of a power converter.

* * * * *